

Four problems, and the scaffolding falls away

The symmetric toggle, with $\alpha = 2$ throughout so that the crossing on the diagonal is $x = 1$ by hand:

$$\frac{du}{dt} = \frac{\alpha}{1 + v^n} - u \qquad \frac{dv}{dt} = \frac{\alpha}{1 + u^n} - v$$

Tuesday's two tools, and nothing else: **fixed points are where the nullclines cross**, and a crossing is a **saddle exactly when $g_1 g_2 > 1$** , with

$$g_1 = \frac{n u v^{n-1}}{1 + v^n}, \quad g_2 = \frac{n v u^{n-1}}{1 + u^n}, \quad \lambda = -1 \pm \sqrt{g_1 g_2}.$$

At each transition the sheet asks *why does that step follow?* Answer in writing before you go on.

1 • Fully worked — $n = 4, \alpha = 2$

The diagonal crossing. On $u = v = x$ both equations read $x + x^5 = 2$, and $x = 1$ solves it: $1 + 1 = 2$.

Is it a saddle? At $u = v = 1$, $g_1 = g_2 = 4 \cdot 1 \cdot 1 / (1 + 1) = 2$, so $g_1 g_2 = 4 > 1$. Yes:

$\lambda = -1 \pm 2 = +1, -3$. The cell does not sit there.

The other two crossings. They are off the diagonal and mirror each other.

`stability_report(toggle_model(2, 2, n=4))` puts them at (0.118, 2.000) and (2.000, 0.118).

Check one by hand: $2/(1 + 2.000^4) = 2/17 = 0.118 \checkmark$ and $2/(1 + 0.118^4) = 2/1.0002 = 2.000 \checkmark$.

► **Why does that step follow?** Why is there always a crossing on the diagonal, and why do the other two come as a mirror pair?

2 • Last step blank — classify the pair

At $(u, v) = (0.118, 2.000)$, with $n = 4$:

$$g_1 = \frac{4 \cdot 0.118 \cdot 2.000^3}{1 + 2.000^4} = \underline{\hspace{2cm}} \qquad g_2 = \frac{4 \cdot 2.000 \cdot 0.118^3}{1 + 0.118^4} = \underline{\hspace{2cm}}$$

► $g_1 g_2 = \underline{\hspace{2cm}}$ so the point is $\underline{\hspace{2cm}}$, with $\lambda = \underline{\hspace{2cm}}$.

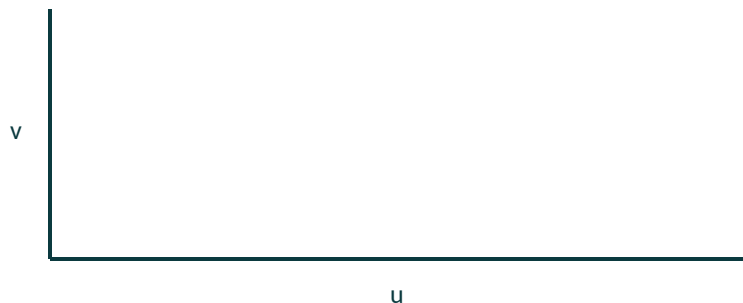
► **Why does that step follow?** One factor comes out roughly twenty times the other. Say in one sentence why the *product* is what decides, and what it means that one factor is so small.

3 · Last two blank — $n = 1, \alpha = 2$

The diagonal crossing. $x + x^2 = 2$, so $x = 1$ again.

► Build g_1 and g_2 there, then classify:

► How many crossings are there in total, and why? Sketch the two nullclines for $n = 1$ on the axes and say what Tuesday's handout proved about this case.



4 · Bare problem — weaken one arm

Back to $n = 4$. Keep $\alpha_1 = 2$ and lower α_2 from 2 toward 1, so the u arm is now the stronger one. Which nullcline moves, and how? Which of the three crossings dies first, what does it die *with*, and what is the cell left in? Draw it; then name the event in the language of today's lecture, and name the plasmid in the paper that did this to itself.

